

Sayed Omar

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ABOUT ME

Final-year Aircraft Manufacturing Engineering student (7th of 8 semesters, graduating June 2027). My research combines piezoelectric energy harvesting on UAV structures with deep reinforcement learning for energy-aware navigation. I built the finite-element and reinforcement-learning simulation framework, and the work has led to two journal manuscripts that are now under review. A key result is negative but useful: arm-mounted harvesting cannot extend flight time, while energy-aware flight can. I am applying for MSc programmes where I can take this work from simulation to bench and in-flight experiments.

EDUCATION AND TRAINING

09/2023 – 06/2027 (EXPECTED) · TIANJIN, CHINA

Bachelor of Engineering in Aircraft Manufacturing Engineering

Tianjin Sino-German University of Applied Sciences (TSGUAS)

- **CGPA:** 3.17 / 5.0 · **Current aggregate:** 82%
- **Thesis (in progress):** “Piezoelectric Energy Harvesting Coupled with Energy-Aware Deep Reinforcement Learning for Extended-Endurance Autonomous UAVs”. Proposal defence rated **Exceptional** by the committee (2026).
- **Relevant coursework:** Structural Mechanics · Finite Element Analysis · Control Systems · Machine Learning · CAD/CAM · Aircraft Manufacturing Processes · Aerodynamics · Materials Science

RESEARCH EXPERIENCE

09/2024 – CURRENT · TSGUAS, TIANJIN, CHINA

Undergraduate Researcher: Piezoelectric Energy Harvesting on Quadrotor Arms (FEA)

Supervisors: Prof. Ma Guoli (co-author), Prof. Liu Guanwei (thesis supervisor)

- Built an Euler–Bernoulli finite-element model of the DJI F450 arm with the motor/propeller tip mass and a surface-bonded PZT-5A patch; verified against closed-form solutions (first-mode error 8.5×10^{-6} %).
- Showed that omitting the motor tip mass (bare-beam model) raises the first bending frequency from 18.4 Hz to 57.7 Hz and overestimates mission-average harvested power by about **28×**.
- A root-mounted patch delivers **0.03 μ W in hover** and 0.18 μ W averaged over a mission; four patches recover ~ 0.42 mJ per 10-min flight, $\approx 3 \times 10^{-9}$ of propulsion energy: suitable only for duty-cycled sensing, not for extending endurance.
- Designed a bench-test protocol (arm + motor + propeller, 3,000–8,200 rpm, strain and patch voltage across a matched load) for experimental validation, planned as next step.

01/2025 – CURRENT · TSGUAS, TIANJIN, CHINA

Undergraduate Researcher: Energy-Aware UAV Navigation via Deep Reinforcement Learning

Supervisor: Prof. Ma Guoli (co-author)

- Implemented DQN, PPO and SAC in NumPy, with rotor-speed-dependent propulsion power and the FE harvest model evaluated at every step.
- Over 5 seeds and 200 identical test scenarios, SAC achieved the highest success (**$83.6 \pm 3.6\%$** , vs 72.4% PPO and 59.5% DQN; ANOVA $p = 0.0035$), while PPO used 15% less energy per metre, a trade-off rather than a single best algorithm.
- An energy term in the reward cut propulsion energy per metre by **13%** without loss of success; a harvest-seeking reward bonus proved counterproductive (+12.6% energy).
- Verified every learning gradient against finite differences (25/25 unit tests); code and raw results are public on GitHub.

09/2023 – 06/2024 · TSGUAS, TIANJIN, CHINA

Independent Project: CATIA–ANSYS Integrated Workflow for Structural Analysis

Supervisor: Prof. Pen Hongmin

- Developed a Python pre-processor converting CATIA parametric geometry into structured FEA meshes with controlled aspect ratios; validated against analytical cantilever-deflection solutions.
- Linked geometry and FEA in both directions, so design changes no longer needed manual re-meshing.
- Used the workflow for stress analysis of aerospace parts; this is where I learned the FEA methods I later used in my thesis.

PUBLICATIONS

Both manuscripts are currently under peer review.

[1] **Can Arm-Mounted Piezoelectric Harvesting Extend Quadrotor Endurance? A Physics-Consistent Finite-Element and Energy-Aware Deep Reinforcement Learning Assessment**

S. Omar, G. Ma · *International Journal of Intelligent Robotics and Applications*, 2026 (under review) · Code and data: github.com/omar123-1-cyber/UAV-Energy-Research

[2] **Piezoelectric Energy Harvesting from Multirotor UAV Structures: A Systematic Review of Placement Optimization, Broadband Strategies, and Pathways to Flight-Ready Systems (2008–2026)**

S. Omar, G. Ma · *Noise & Vibration Worldwide*, 2026 (under review) · Manuscript ID: NVW-26-0052

LANGUAGE SKILLS

Mother tongue: Bengali

Other languages	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	B2	C1	C1	C1	B2
Mandarin Chinese	B1	A2	B1	B1	A2
Arabic	B1	B1	B1	B1	A2
Urdu	B1	A2	B1	B1	A2

Levels: A1/A2 basic user; B1/B2 independent user; C1/C2 proficient user (CEFR). English: 4-year English-medium-of-instruction certificate; IELTS scheduled for November 2026.

DIGITAL SKILLS

Programming	Python (NumPy, SciPy, Matplotlib), MATLAB, JavaScript (Next.js, React, Express.js), LaTeX; full-stack web development
Databases & tools	MongoDB, Jupyter, Linux (Ubuntu), Git/GitHub, VS Code; all research code version-controlled, manuscripts written in LaTeX
Machine learning & AI	DQN, PPO, SAC implemented in NumPy; statistical analysis (ANOVA, Welch t-test, Holm correction, effect sizes)
CAD / FEA	CATIA V5/V6, SolidWorks, ANSYS Mechanical (structural FEA, meshing); Python Euler–Bernoulli FE
Robotics	ABB RobotStudio
Hardware & embedded	Raspberry Pi, Arduino

HONOURS AND AWARDS

2026 · TSGUAS

Exceptional rating, thesis proposal defence

HOBBIES AND INTERESTS

Drones & autonomy	UAV energy efficiency and autonomous navigation.
Open-source software	Web development (Next.js, React, Express.js) and small scientific computing tools, shared on GitHub.
Competitive programming	Algorithm problems, data structures and numerical methods.

REFERENCES

Prof. Ma Guoli, Research supervisor and co-author, TSGUAS, Tianjin, China · bz_mgl@163.com

Prof. Liu Guanwei, Thesis supervisor, Dept. of Aircraft Manufacturing Engineering, TSGUAS / Tianjin University · tjulgw@tju.edu.cn

Prof. Pen Hongmin, Project supervisor, TSGUAS, Tianjin, China · hmpen23@126.com